Ram Pandey

Ph.D. Candidate in Electrical Engineering

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RESEARCH INTERESTS

Power system **reliability, stability & resiliency** · Microgrid operation & control · Distributed energy resources · Real-time power system simulation · AI/ML applications in power systems · Frequency security & virtual synchronous machines

EDUCATION

Ph.D. in Electrical Engineering (Ongoing) 2023 – 2027 (expected)
University of Wyoming · *Laramie, WY, USA*

M.E. in Electrical Power Engineering — *Outstanding Student Award* 2015 – 2017
Kathmandu University · *Dhulikhel, Nepal*

B.E. in Electrical Engineering — *Government Full Scholarship* 2010 – 2014
Tribhuvan University, Pashchimanchal Campus · *Nepal*

PROFESSIONAL EXPERIENCE

Graduate Research Assistant Aug 2023 – Present
University of Wyoming · *Laramie, WY, USA*

- Conducting research aimed at improving the reliability, optimal scheduling, and resilience of power systems that rely heavily on stochastic and low-inertia generation and variable loads.
- Published **5 peer-reviewed conference papers** since 2024 at flagship IEEE venues (TPEC, NAPS, ETFG) and have submitted a few journal papers.

Research Collaborator May 2025 – Aug 2025
Sandia National Laboratories · *Albuquerque, NM, USA*

Assistant Professor Nov 2017 – Jul 2023
Tribhuvan University, Institute of Engineering · *Nepal*

- Served as **Head of Department** (Apr 2021 – Jul 2023): led academic programme management, faculty coordination, and curriculum development for the Electrical Engineering department.
- Taught 12+ courses across power systems, power electronics, control systems, and renewable energy.
- Supervised undergraduate and postgraduate research; co-authored multiple peer-reviewed publications with students.

Lecturer Aug 2016 – Oct 2017
National College of Engineering, Tribhuvan University · *Nepal*

Teaching Assistant Sep 2015 – Jul 2016
Kathmandu University · *Dhulikhel, Nepal*

Project Engineer (Electrical) May 2015 – Aug 2015
Technical Engineering Design Consultancy & Construction Pvt. Ltd. · *Kathmandu, Nepal*

- Prepared a detailed feasibility study for a rural micro-hydropower project in western Nepal, including hydrological, geographical, technical, and economic assessments.

- Designed complete rural power evacuation infrastructure, including distribution lines, transmission systems, and load-serving electrical networks.
- Engineered a small-scale rural substation with transformer sizing, protection coordination, grounding, and reliable system integration for community electrification.

Teaching Assistant Dec 2014 – Apr 2015
Thapathali Campus, IOE, Tribhuvan University · *Nepal*

PUBLICATIONS

1. **R. P. Pandey** and N. Nguyen, "Scenario-Aware Scheduling in Microgrids: Look-Ahead Optimization with Frequency Security," *2025 IEEE Texas Power and Energy Conference (TPEC)*, pp. 1–6, IEEE, 2026.
2. **R. P. Pandey** and N. Nguyen, "Frequency Secure Optimal Scheduling of Microgrids with Energy Storage Support," *57th North American Power Symposium (NAPS)*, IEEE, 2025.
3. **R. P. Pandey** and N. Nguyen, "Electric Vehicle-Based Frequency Support in Low-Inertia Power Grids," *IEEE International Conference on Energy Technologies for Future Grids (ETFG)*, Wollongong, Australia, Dec. 2025.
4. **R. P. Pandey** and N. Nguyen, "Reliability Assessment of VSM-Based Microgrid Incorporating Transient Stability," *2025 IEEE Texas Power and Energy Conference (TPEC)*, pp. 1–6, IEEE, 2025.
5. **R. P. Pandey**, U. Tamrakar, and N. Nguyen, "Machine Learning-Based Composite Reliability Assessment of Power Systems," *56th North American Power Symposium (NAPS)*, pp. 1–6, IEEE, 2024.
6. **R. P. Pandey** and S. Gurung, "Small Signal Stability Analysis with Wind Power Penetration in the INPS System," *International Journal of Current Research*, vol. 13, no. 1, pp. 15452–15456, 2021.
7. **R. P. Pandey**, "Study of Small Signal Stability Using PSAT Including DFIG-Based Wind Penetration," *Journal of Innovation in Engineering Education*, vol. 2, no. 1, 2019.
8. B. Jha, A. Adhikari, and **R. P. Pandey**, "Design of Controlled Charging Strategy for Parallel Operation of Multiple Modular Lithium-Ion Battery Packs," *2nd International Conference on Sustainable Mobility Applications, Renewables and Technology (SMART)*, pp. 1–8, IEEE, 2022.
9. B. Jha, B. Mallik, S. Basnet, A. Yadav, and **R. P. Pandey**, "Design and Simulation of Different Variants of Charging Lithium-Ion Batteries in Search for Optimum Charging Algorithm," *4th International Conference on Smart Systems and Inventive Technology (ICSSIT)*, pp. 752–759, IEEE, 2022.
10. B. Basnet, K. Sapkota, R. Poudel, S. Kshetri, and **R. P. Pandey**, "Dynamic-Static VAR Compensation for Improving Power Factor," *Journal of Engineering and Sciences*, vol. 1, no. 1, pp. 46–49, 2022.
11. A. Ghimire *et al.* and **R. P. Pandey**, "Continuous Supply to Load by Automatic Switchover During Fault Using PLC and SCADA," *Journal of Engineering and Sciences*, vol. 1, no. 1, 2022.
12. B. Bimali, S. Uprety, and **R. P. Pandey**, "VAR Compensation on Load Side Using Thyristor Switched Capacitor and Thyristor-Controlled Reactor," *Journal of the Institute of Engineering*, vol. 16, no. 1, 2021.
13. S. B. Chhetria, **R. P. Pandey**, and M. Karki, "Optimal Conductor Reinforcement and Hosting Capacity Enhancement for Radial Distribution Line of Beni Feeder," *10th IOE Graduate Conference*, 2021.
14. S. P. Upadhyay, **R. P. Pandey**, and M. Karki, "Cost of EENS and Reliability Worth Analysis: A Case Study of Pokhara Industrial Estate," *10th IOE Graduate Conference*, 2021.
15. T. Giri, **R. P. Pandey**, S. Bhandari, S. Moktan, and L. Karki, "Design and Simulation of Three-Phase Three-Wire Shunt Active Filter Using Instantaneous PQ Theory," *Journal of the Institute of Engineering*, vol. 15, no. 1, pp. 181–186, 2019.

TECHNICAL SKILLS

Simulation	OpenDSS, MATPOWER, Simulink, PSAT, DIgSILENT PowerFactory, RT-Lab, ePHASORSIM, PyPSA, pandapower, Electrical AutoCAD
Programming	Python, MATLAB, DPL, C, C++
ML / Scientific	PyTorch, TensorFlow, Scikit-learn, Pyomo, NumPy, Pandas
Documentation	LaTeX, MS Office Suite

TRAINING & WORKSHOPS

Sep 2024	**ePHASORSIM – Electro-Mechanical Real-Time Simulation**, OPAL-RT Technologies
Sep 2024	**RT-Lab Real-Time Simulation Systems Fundamentals**, OPAL-RT Technologies
Apr 2022	**Comprehensive Training in DIgSILENT PowerFactory and DPL**, Pashchimanchal Campus, IOE, TU
Dec 2019	**Three-Week Energy Audit Training**, Nepal Energy Efficiency Program (NEEP), GIZ & IOE, TU
Nov 2019	**Lightning Protection on Structures and Electrical Networks**, South Asian Lightning Network
Jun 2019	**Seven-Day High Voltage Testing Training**, Pulchowk Campus, IOE, TU

COURSES TAUGHT

- Distributed Energy Resources
- Distribution System Planning & Management
- Power System Optimization & Control
- Flexible AC Transmission Systems (FACTS)
- Advanced Photovoltaic Systems
- Digital Simulation & Analysis of Power Systems
- Energy Audit, Analysis & Conservation
- High Voltage Engineering
- Control Systems
- Utilization of Electrical Energy
- Power Electronics
- Basic Electrical Engineering

AWARDS & HONOURS

2025–2026	**School of Computing, Graduate Fellowship**, School of Computing, University of Wyoming
2017	**Outstanding Student Award**, Kathmandu University
2010–2014	**Government Full Scholarship** for undergraduate study

SERVICE & LEADERSHIP

2021–2023	**Head of Department**, Electrical Engineering, Pashchimanchal Campus, Institute of Engineering, Tribhuvan University, Nepal
2023–present	**IEEE Student Member**
2013–2014	**President**, Electro-Genesis Club, Pashchimanchal Campus
2014–2015	**President**, Janachetana Youth Club, Gorkha, Nepal